

Luke Hyatt

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EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC

BS in Computer Science, Math | GPA: 3.6/4.0 | Expected Graduation: May 2028

Aug. 2024 – May. 2028

Relevant Coursework: Algorithms and Analysis, Real Analysis, Data Structures, System Fundamentals, Discrete Math, Differential Equations, Linear Algebra, Probability

TECHNICAL SKILLS

Languages: Java, Python, C, C++, Go, Swift, SQL, HTML/CSS

Frameworks & Libraries: React, TensorFlow, PyTorch, pytest, NumPy, pandas, Matplotlib

Developer Tools: Git, Docker, VSCode, PyCharm, Xcode, Arduino, Jupyter, Linux/bash

Concepts & Methodologies: OOP, REST APIs, Agile, CI/CD, unit testing, agentic workflow

Databases: Oracle, SQL Server, Snowflake, Teradata, SingleStore

PROFESSIONAL EXPERIENCE

Software Engineer Intern

May 2026 - Aug 2026

SAS

Cary, NC

- Increased automated Python test coverage of SAS Studio Micro Flow services by 50% through developing 134 integration tests across five database backends (Oracle, SQL Server, Snowflake, Teradata, SingleStore)
- Created parameterized pytest suites for 2 Micro Flow data flow steps, covering 102 and 32 configuration permutations respectively

ML Research Assistant

Oct 2025 – May 2026

UNC Computational Robotics Lab

Chapel Hill, NC

- Built the benchmarking pipeline that compared the proposed model against prior work across all evaluation scenarios, producing the head-to-head results reported in a paper under review at ICRA 2027
- Developed 2D and 3D spherical simulation scenarios and Python visualization tooling that generated figures used in the submission

Research Assistant

Jan 2026 – Current

UNC Experimental Engineering Lab

Chapel Hill, NC

- Architected an autonomous navigation system for a Boston Dynamics Spot robot to act as a guide dog for the visually impaired, routing user-entered destinations over a prerecorded waypoint graph of Chapel Hill
- Prototyped the destination-to-navigation pipeline in Python, mapping mobile app input to localized waypoints and issuing movement commands to the robot through the Spot SDK

Teaching Assistant- System Fundamentals Course

Jan 2026 – May 2026

University of North Carolina at Chapel Hill

Chapel Hill, NC

- Supported 153 students in a systems fundamentals course through 10 weekly office hours on C programming, git, and computer architecture
- Assisted instruction for an 80-student lecture section, graded 86 assignments and 5 exams

PUBLICATIONS

Anatomy-Aware Dexterity-Driven Design Optimization of Surgical Continuum Robots

2027

Under review, IEEE Int. Conf. on Robotics and Automation (ICRA)

- T. Qin et al. (**co-author**). Contributed benchmark scenarios, a model-vs-prior-work comparison pipeline, and figures used in the paper.

PROJECTS

Network Diagnostic App | *Swift, C, Xcode, REST API, Git*

May 2025 – Aug 2025

- Developed an iOS App for student iPads for K-12 Technology Department
- Diagnoses network problems on school iPads and sends critical information to Tech department via POST requests

Posture Checker | *Python, TensorFlow, PyCharm, Git*

May 2025 – Aug 2025

- Built a computer-vision model that detects slouching in real time from webcam input
- Extracted body keypoints with TensorFlow MoveNet Lightning and classified posture from spine angle